



Leveraging Robotics and AI in Surveillance



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Motivation

Robotics and Artificial Intelligence (AI) can revolutionize tasks that are Dirty, Dull, and Dangerous—areas where we hold substantial expertise. These technologies are particularly impactful in environments requiring intensive labor and continuous monitoring.

Problem Space

Surveillance of linear infrastructures like pipelines and waterways is labor-intensive, risky, and monotonous. These tasks tie up valuable manpower and expose personnel to hazardous conditions.

Three D's of Robotics

- Dirty:** Environments that threaten human health.
- Dull:** Repetitive tasks prone to human error.
- Dangerous:** Conditions risky to human safety.

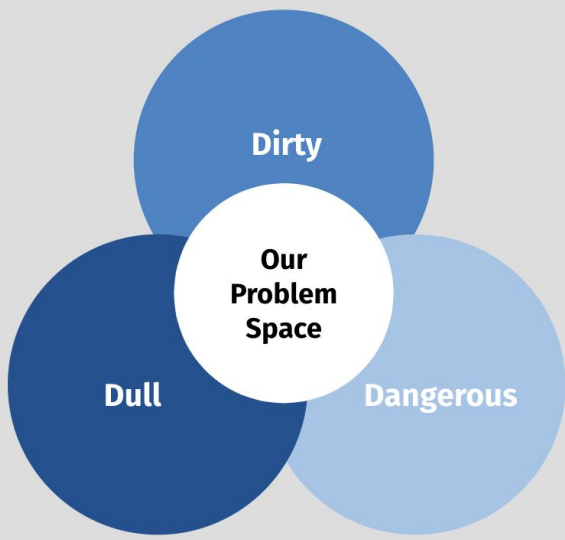


Figure 1: Three D's of Robotics

Focus

Stationary Surveillance: Current stationary surveillance systems for monitoring infrastructures can greatly benefit from robotic automation and AI analytics.

Our Investigation

Can robotics and AI automate surveillance to free up manpower for critical tasks?

Proposed Solution : Spy Turtle

Goal

Enhance surveillance capabilities through advanced robotics and AI, focusing on human tracking and autonomy.

Key Features

- Computer Vision:** Leverage AI to distinguish between 'enemy' and 'friendly' classifications based on clothing color.
- Robotics & Networking:** Utilize hardware controlled with ROS 2 for robust robotic systems with modular functionality.
- Human-Machine Interaction:** Implement interfaces for real-time human interaction and control adjustments via a web application.

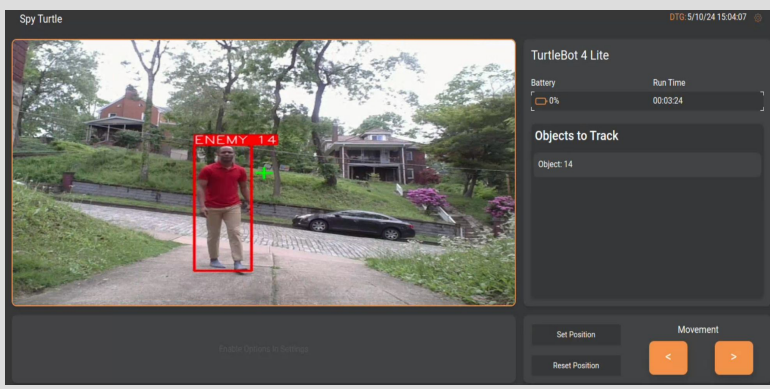


Figure 2: Web App Interface

Technology Stack

- Robotics:** TurtleBot 4 Lite, equipped with necessary sensors (camera, LIDAR) for object detection.
- Artificial Intelligence:** Custom-trained model using YOLOv8n for real-time enemy and friendly class detection.
- Networking:** ROS 2 to handle communication within the different robotic functions and SocketIO for communication between the ROS2 system and the web server.
- Web Technologies:** Flask application serving a frontend created with HTML, CSS, and JavaScript, enabling user interactions and live feed display.

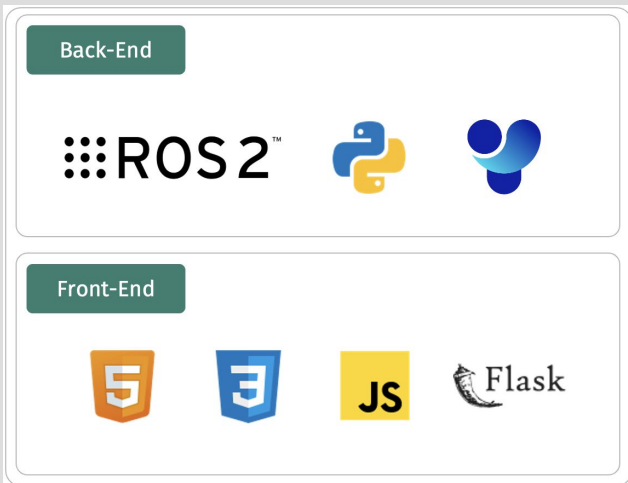


Figure 3: Web App Technology Stack

System Architecture

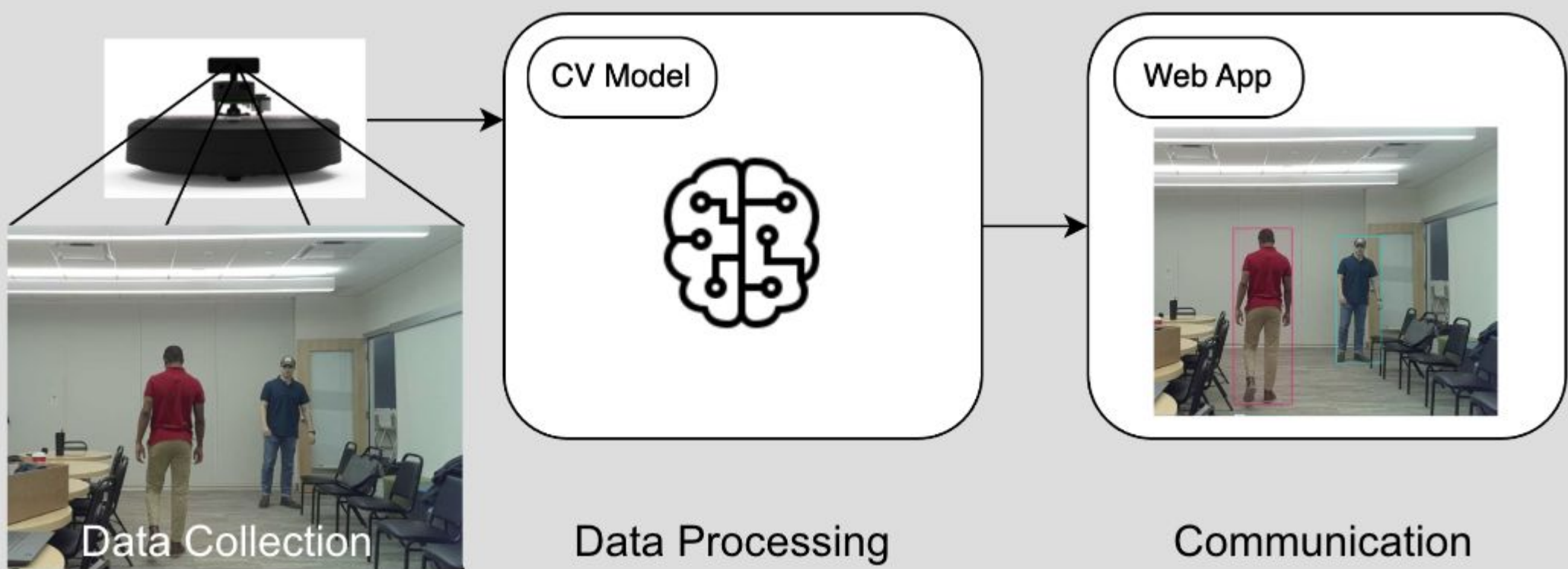


Figure 4: This table showcases the data flow architecture of the Spy Turtle application

Results

Metric	Baseline YOLOv8n	Fine-Tuned YOLOv8n	Improvement
F1 Score	0.11	.85	+672.73%
mAP50-95	0.13	0.71	+446.15%

Figure 5: This table showcases the substantial enhancements achieved through fine-tuning the YOLOv8n model on our specific dataset.



Figure 6: These images show the friendly/enemy classes and the object tracking

Analysis and discussion of results

Results Interpretation

The significant improvements in F1 Score (+672.73%) and mAP50-95 (+446.15%) enhance the YOLOv8n model's accuracy and reliability in distinguishing 'enemy' and 'friendly' targets. These metrics underscore a marked advancement in precision and recall, crucial for dependable surveillance operations.

Impact on Project Goals

These advancements align directly with the "Spy Turtle" objectives of improving surveillance effectiveness and autonomous capabilities. The results validate our methodological choices and fine-tuning of YOLOv8n, demonstrating substantial progress toward robust human-machine collaboration.

Limitations

While results are promising, they are contingent on controlled environments and specific data types. Performance variations under different conditions and with unrepresented data warrant further investigation.

Future Work

To broaden model robustness and environmental adaptability, we plan to utilize more diverse data sets. Advanced AI algorithms will be tested, and hardware enhancements investigated to boost real-time data processing capabilities. Additionally, we aim to optimize system functionality for limited network environments by improving onboard computing and refining data streaming processes, ensuring consistent performance under varying connectivity conditions.

Key Takeaways

Significant Performance Gains

The fine-tuning of the YOLOv8n model led to dramatic improvements: a 672.73% increase in the F1 Score and a 446.15% rise in mAP50-95, showcasing enhanced precision in detecting and differentiating between 'enemy' and 'friendly' targets.

Enhanced Surveillance Capabilities

The project demonstrates a viable application of advanced robotics and AI in surveillance, significantly improving the autonomy and accuracy of the TurtleBot in real-world scenarios.

Real-World Application

"Spy Turtle" illustrates how integrating AI with robotics can reduce the need for human intervention, making surveillance tasks more efficient and safer, particularly in environments where human presence is risky or impractical.

Limitations and Future Potential

Despite current successes, challenges remain, particularly in adapting the system to diverse and dynamic environments. Ongoing and future developments will focus on enhancing the robustness and adaptability of the system.